Rishon Ghosh

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Education

Simon Fraser University — ***Burnaby, BC, Canada***

Bachelor of Science, Computer Science (Joint Major with Business) | ***GPA: 3.89*** — **Expected Grad Date: Dec 2028**

- **Coursework**: Algorithms & Data Structures, Introduction to Software Engineering, Business Technology Management
- **Awards**: Dean's Honour Roll, President's Honour Roll, SFU Undergraduate Open Scholarship

Experience

Software Engineer Consultant — ***Vancouver, BC***

RG Tech — **Jul 2025 - Present**

- Architected Python and Flask REST backends for **3 small-business clients**, turning existing sales and inventory data into reports
- Engineered SQL schemas and ETL scripts consolidating spreadsheet and POS data, saving **12 hours** of manual entry weekly
- Optimized database performance through indexing and query refactoring, reducing average API response times by **40%**
- Implemented secure authentication with password hashing and input validation to protect client and customer records
- Executed client discovery, scoping, and sprint planning to translate business needs into technical requirements delivered on schedule

Projects

MatchMesh | *C++20, gRPC, Protocol Buffers, CMake, GoogleTest, Docker, Angular* — **Oct 2026**

- Built a fault-tolerant matchmaking cluster of **5** C++20 nodes over gRPC, sharding queues on a consistent hash ring so a node failure remaps only **∼17%** of queues versus **∼80%** with modulo hashing
- Engineered synchronous primary-backup replication with per-queue sequence numbers and epoch-fenced writes, plus heartbeat failure detection that fails a killed node's queues over in **1.3–1.5 s**
- Wrote a chaos harness that kills nodes under load and an invariant checker: **0** players lost or double-matched across **11** kills and **79,000** players, after it exposed and drove fixes for **2** real bugs
- Load-tested open-loop to **0.78 ms** p50 and **3.07 ms** p99 enqueue latency at 1,000 players/s; lock striping doubled throughput, verified by **231** tests (**165** under ThreadSanitizer)

TrustLine | *TypeScript, React, Node.js, ElevenLabs, Gemini, Twilio, Supabase, Docker* — **Oct 2026**

- Architected a real-time audio intelligence pipeline streaming live phone calls from Twilio Media Streams to ElevenLabs Scribe v2 Realtime over WebSockets, transcribing **8 kHz** telephony audio with no transcoding and warning callers mid-call in **5 languages**
- Engineered a dual-layer scam detection engine fusing a **∼0.01 ms** rules layer across **12 scam tactics** with schema-constrained, Zod-validated LLM scoring (Google Gemini), achieving **100% precision and recall** with **0 false positives** on a 17-call labelled set at **1.1 s** median latency
- Established an evaluation and CI framework measuring precision, recall and alert latency, gating every push through GitHub Actions with **187 automated tests** across **28 suites**
- Orchestrated production deployment on **$0** infrastructure: a Dockerized call server on Railway with health-checked, path-filtered auto-deploys, and a React/Vite frontend with serverless APIs on Vercel

FlashCart | *Go, TypeScript, React, AWS Lambda, DynamoDB, EventBridge, SQS, CDK* — **Sep 2026**

- Built a serverless flash-sale checkout in Go on AWS Lambda and API Gateway that sustained **∼6,900 req/s** at **1,000** concurrent k6 users with **132 ms** average and **194 ms** p95 latency
- Prevented overselling by committing the stock decrement and order insert in one DynamoDB transaction keyed on the client's idempotency key: **0** oversold units across **145,000+** requests, backed by a **500**-goroutine race test
- Removed dual writes with a transactional outbox (DynamoDB Streams to EventBridge Pipes to SQS), an idempotent worker with compensating refunds, and a dead-letter queue with CloudWatch alarms
- Defined the stack in AWS CDK with template assertion tests; GitHub Actions runs Go tests against DynamoDB Local and deploys through keyless OIDC on every push to main

Graphite Gambit | *Java, libGDX, JUnit, Mockito, Maven, JaCoCo* — **Jan 2026**

- Led a team building a 2D top-down game in Java and libGDX, establishing weekly agile sprints and a merge-conflict protocol
- Created a map-loading system and collision engine, using mathematical proofs in testing to verify safe entity movement and bounds
- Automated a testing suite with JUnit and Mockito, achieving **71%** branch coverage across core game packages

Technical Skills

Languages: Go, C++, TypeScript, Java, Python, SQL, C, JavaScript, HTML/CSS
Frameworks & Libraries: React, Angular, Node.js, Flask, gRPC, Protocol Buffers, Zod, libGDX, JUnit, Mockito, GoogleTest
Cloud & DevOps: AWS (Lambda, API Gateway, DynamoDB, SQS, EventBridge, S3, CloudFront, CDK), Docker, GitHub Actions
Databases: DynamoDB, PostgreSQL (Supabase), SQLite
Developer Tools: Git, CMake, Maven, k6, JaCoCo, ThreadSanitizer, VS Code
Concepts: Distributed Systems, Fault Tolerance, Consistent Hashing, Concurrency, Idempotency, Event-Driven Architecture, CI/CD
Business: Financial Accounting, Excel, PowerPoint, Sprint Planning & Execution